

DANIEL MANCILLA TEJERINA

Fullstack & Backend Software Engineer | .NET · C# · Python | Azure Cloud & DevOps | Clean Architecture · Microservices · CQRS · REST APIs | React · Angular · React Native | AI Agents | 2x ACM-ICPC Regional Finalist

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Languages: Spanish (Native) | English (B1 Technical / Intermediate)

PROFESSIONAL SUMMARY

Systems Engineer with +1 year of experience developing fullstack web applications, microservices, and scalable backend systems using modern technologies (.NET / C#, Python) with cloud deployments on Azure. Experienced in designing and implementing decoupled enterprise architectures applying patterns such as Clean Architecture, CQRS, and designing secure RESTful APIs with ASP.NET Core, as well as offline-first mobile applications.

Experienced in deploying cloud solutions with Microsoft Azure (App Services, Azure SQL, Blob Storage), container management with Docker, and CI/CD workflow automation using GitHub Actions. In the data tier, proficient in modeling, indexing, and optimizing relational and geospatial databases with PostgreSQL (+ PostGIS) and SQL Server. On the frontend and mobile side, experienced in building reactive, cross-platform interfaces with React, Angular, and React Native (Expo). Additionally, actively integrating advanced workflows with Artificial Intelligence Agents (Claude Code, Antigravity, OpenAI Codex, Context & Prompt Engineering) to maximize productivity and solve complex engineering challenges.

Technical and analytical foundation backed by being a 2x Regional Finalist in the ACM-ICPC South America (2024 & 2025) and Top 6 in Bolivia in IEEEExtreme 18.0, enabling the rapid and precise resolution of algorithmic, performance, and concurrency challenges. Dedicated to contributing to high-impact, high-performance software engineering within agile teams driven by technical excellence.

CORE SKILLS

- **Languages:** C#, Python, TypeScript, JavaScript, C++, SQL, PHP.
 - **Backend & Architecture:** C# / .NET 10, ASP.NET Core Web API, Clean Architecture, Microservices, API Gateway, Hexagonal Architecture, CQRS, RESTful APIs, GoF Design Patterns, JWT & RBAC.
 - **Cloud & DevOps:** Microsoft Azure (App Services, Azure SQL, Blob Storage), Docker, Git/GitHub (CI/CD Pipelines, GitHub Actions, Branch Protection), Linux, Postman.
 - **Frontend & Mobile:** React, Angular, React Native (Expo SDK), Expo Router, Tailwind CSS, HTML5, CSS3, Feature-Sliced Design (FSD).
 - **Databases:** PostgreSQL (PostGIS), SQL Server, MySQL, Supabase, Firebase Firestore. Advanced relational data modeling, indexing, and geospatial queries.
 - **AI & Software Agents:** Artificial Intelligence Agents (Claude Code, Antigravity, OpenAI Codex, LLMs), Context & Prompt Engineering, AI API Integration, and Workflow Automation.
 - **Methodologies & Collaboration:** Scrum, Agile, OpenAPI / Swagger Contracts, Code Reviews, Pair Programming, Advanced Algorithmic Problem Solving.
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PROFESSIONAL EXPERIENCE & TECHNICAL PROJECTS

Backend Software Developer | IO Gama Construcciones

Construction Cost & Resource ERP — Microservices for Parametric Budgeting & Cost Control

Tarija, Bolivia | December 2025 - May 2026 (6 months)

- **Microservices Architecture:** Designed the decoupled enterprise system architecture using .NET 10 Web API following Clean Architecture principles, implementing a perimeter API Gateway and autonomous microservices with JWT and RBAC authentication.
- **Parametric Calculation Engine:** Implemented the core estimation engine to compute direct and indirect construction costs (materials, labor yields, social charges, taxes, and margins) with persistence in PostgreSQL.
- **Design & Technical Coordination:** Defined API integration contracts using OpenAPI / Swagger, ensuring standardized communication between the frontend (React) and backend services, while coordinating technical requirements and project tasks.

Creator & Fullstack Mobile Architect | "Lugares Ocultos de Tarija"

Offline-First Mobile App for Rural Trails, Ecotourism, and Local Gastronomy

December 2025 - Present

- **Offline-First Architecture:** Designed and built the mobile app in React Native and Expo SDK 54, integrating MapLibre GL with locally cached vector tiles to enable 100% offline navigation in mountainous areas without cellular coverage.
- **Geospatial Backend:** Configured data synchronization with Supabase and PostgreSQL + PostGIS for spatial queries on hiking trails, viewpoints, and hidden locations not mapped on standard commercial platforms.
- **Engineering & Modularity:** Architected the codebase using Feature-Sliced Design (FSD) and Atomic Design, delivering critical modules such as the SOS emergency coordinate beacon and the "Pasaporte Chapaco" cultural exploration gamification system.

Fullstack Developer & IT Support (Internship) | Cybercorp S.R.L.

Cybercorp Task Management — Real-Time Operations Platform & Live Service Management

Tarija, Bolivia | 2024 - 2025

- **Real-Time Operations Management:** Developed a web platform using React and TypeScript with Firebase Firestore in real time for assigning and tracking technical ticket lifecycles (Pending, En Route, In Progress, Completed), eliminating 100% of manual paper reports.
- **Geolocation & Search:** Integrated map services for field technician dispatch and route optimization, alongside an instant customer lookup engine with consolidated histories of inspections, quotes, and work orders.
- **Network & CCTV Infrastructure:** Planned and installed IP/CCTV video surveillance systems with local TCP/IP networking, subnet routing, and secure remote access.

Computerized Information Systems Technician | Departmental Electoral Tribunal (TED Tarija)

Tarija, Bolivia | August 2025

- Operated and administered the computerized validation system for real-time voter verification during the 2025 National Elections (ODS N° 424/2025), ensuring zero-error margins across electoral records.
- Delivered first-level technical support, resolving citizen registration discrepancies and handling special escalations under strict electoral compliance.

Game & Logic Developer | "Dino Game" (EZ Team)

June 2024 - August 2024

- Developed a 2D/3D Android game using Unity and C#, applying SOLID principles and Object-Oriented Programming (OOP).
 - Optimized mobile performance, focusing on memory allocation, collision physics, and consistent 60 FPS rendering.
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COMPETITIVE ACHIEVEMENTS & PROBLEM SOLVING

- **2x Latin American Regional Finalist in ACM-ICPC (South America / South Finals):**
 - **ICPC 2025 (RISE Team - UPDS Tarija):** Regional finals qualifier; Top 20 nationally in Bolivia. Solved complex algorithmic problems (graphs, dynamic programming, number theory).
 - **ICPC 2024 (DijkstraDOS Team - UAJMS):** Finalist in the South American regional site.
 - **Top 6 in Bolivia — IEEEExtreme 18.0 (UPDS Tarija):** Outstanding national rank in the 24-hour continuous global competitive programming virtual competition (IEEE) [Official Certificate].
 - **Active Competitor on Global Platforms:** Continuous algorithmic problem-solving on Codeforces, AtCoder, LeetCode, and Competitive Programming Network (RPC).
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EDUCATION

- **Universidad Privada Domingo Savio (UPDS)** — B.S. in Systems Engineering (2021 - 2026)
 - **Universidad Autónoma Juan Misael Saracho (UAJMS)** — B.A. in Modern Languages and Tourism (In progress)
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CERTIFICATIONS & CONTINUING EDUCATION

- **Top 6 in Bolivia — IEEEExtreme 18.0 Programming Competition** — IEEE / UPDS (Oct. 2024, Official Certificate).
- **Artificial Intelligence, Arduino & Python Program** — Jeju National University (South Korea) / NIA-WFK (Aug. 2022, 40 hrs).
- **International Technology and Cybersecurity Congress** — AGETIC / UAJMS (Nov. 2024).
- **Google I/O Extended Tarija 2025** — Google Developer Groups (GDG) / Women Techmakers.